



Resistor Value Classifier

AI-powered classification system that identifies resistor types and values using computer vision

Team: The Ohmies — Jaleel Khan, Clay Farley, Andrew Rein, Ethan Messier

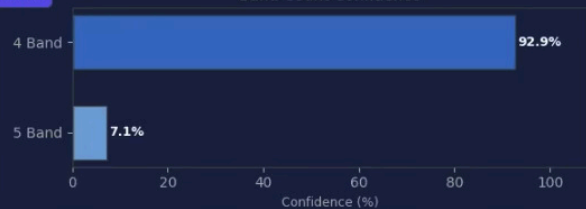


Step 1 — Band Count

4 Band — 92.9% confident

Band Count Confidence

Band Count Confidence

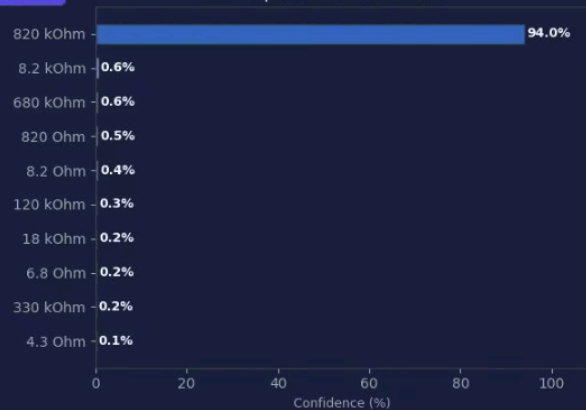


Step 2 — Resistance Value

820 kOhm — 94.0% confident

Value Confidence (Top 10)

Top 10 Value Predictions



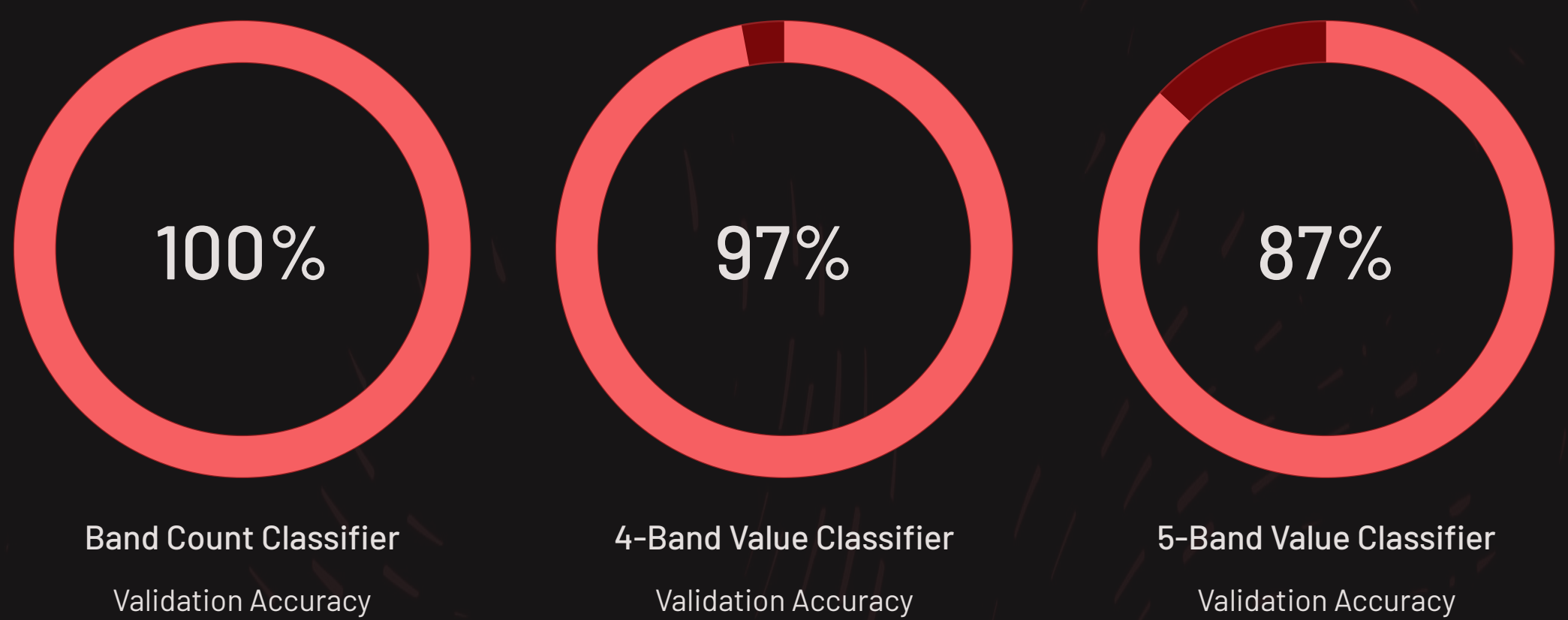
Predicted Color Bands

820 kOhm (4-band)

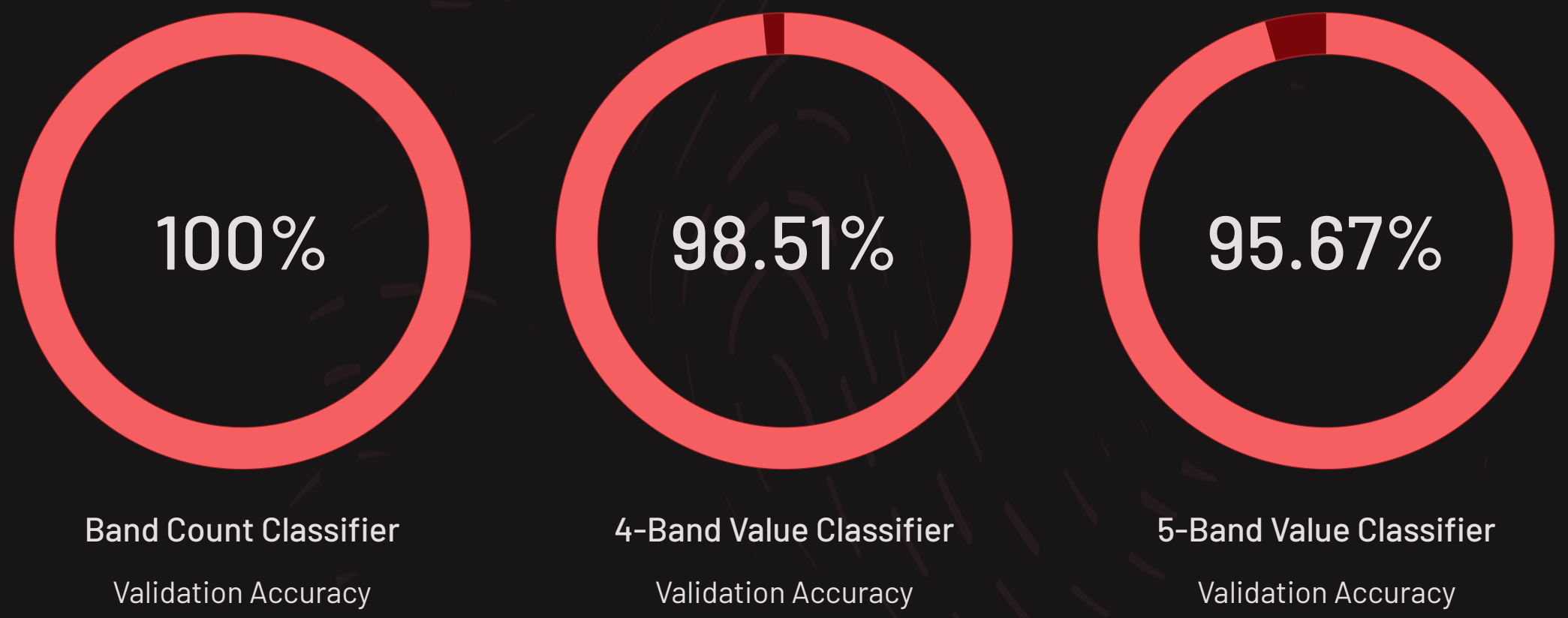


Training Results - ResNet 18

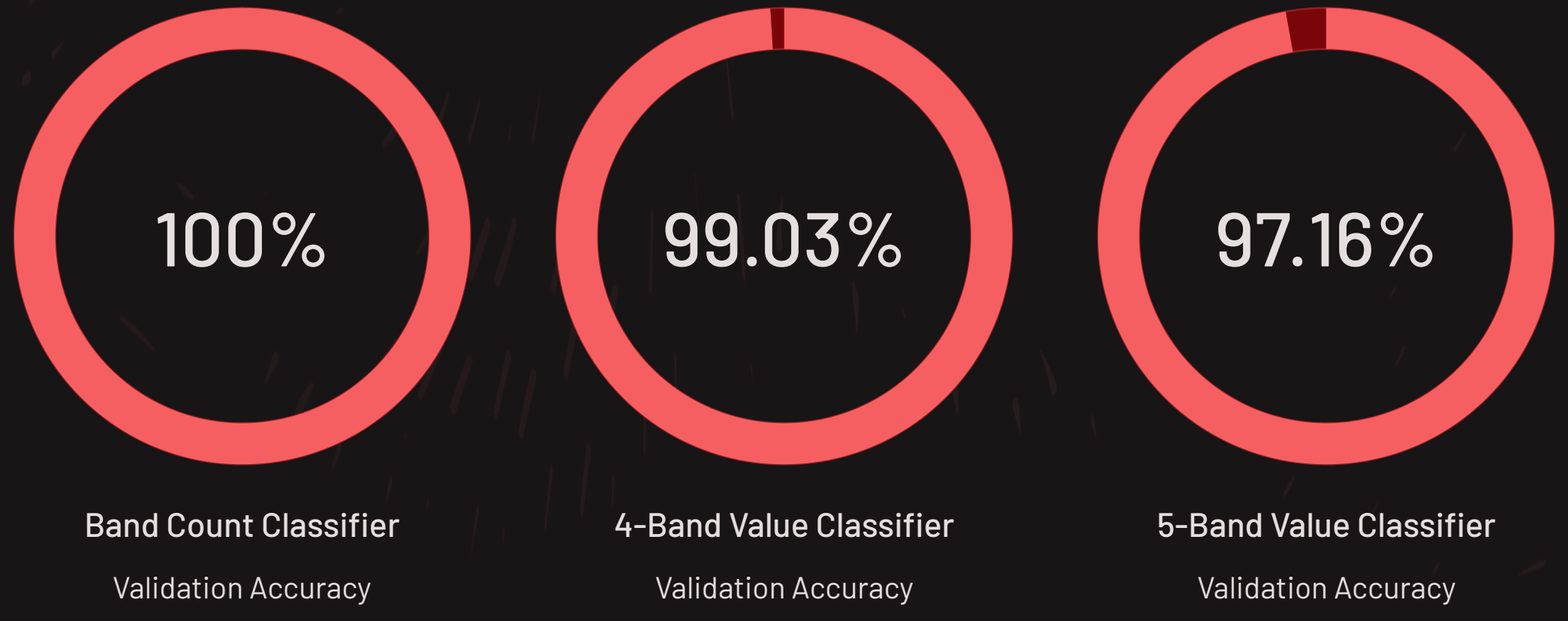
Prototype (~1,200 images)



V1 (~4,000 images)



Final (~15,000 images)



Overall Performance

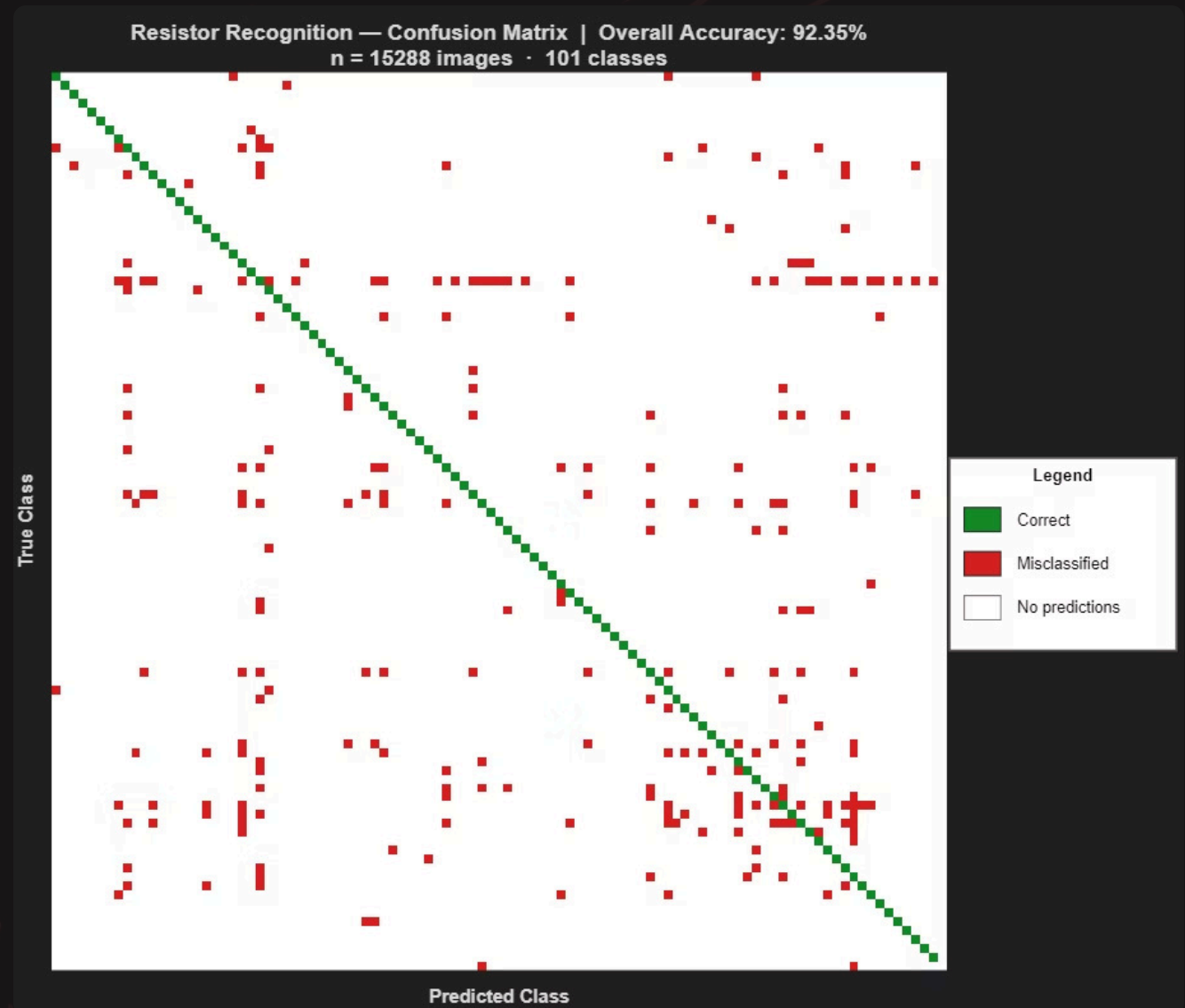
ResNet 18

Model Accuracy Across All Classes

92.35%
accuracy

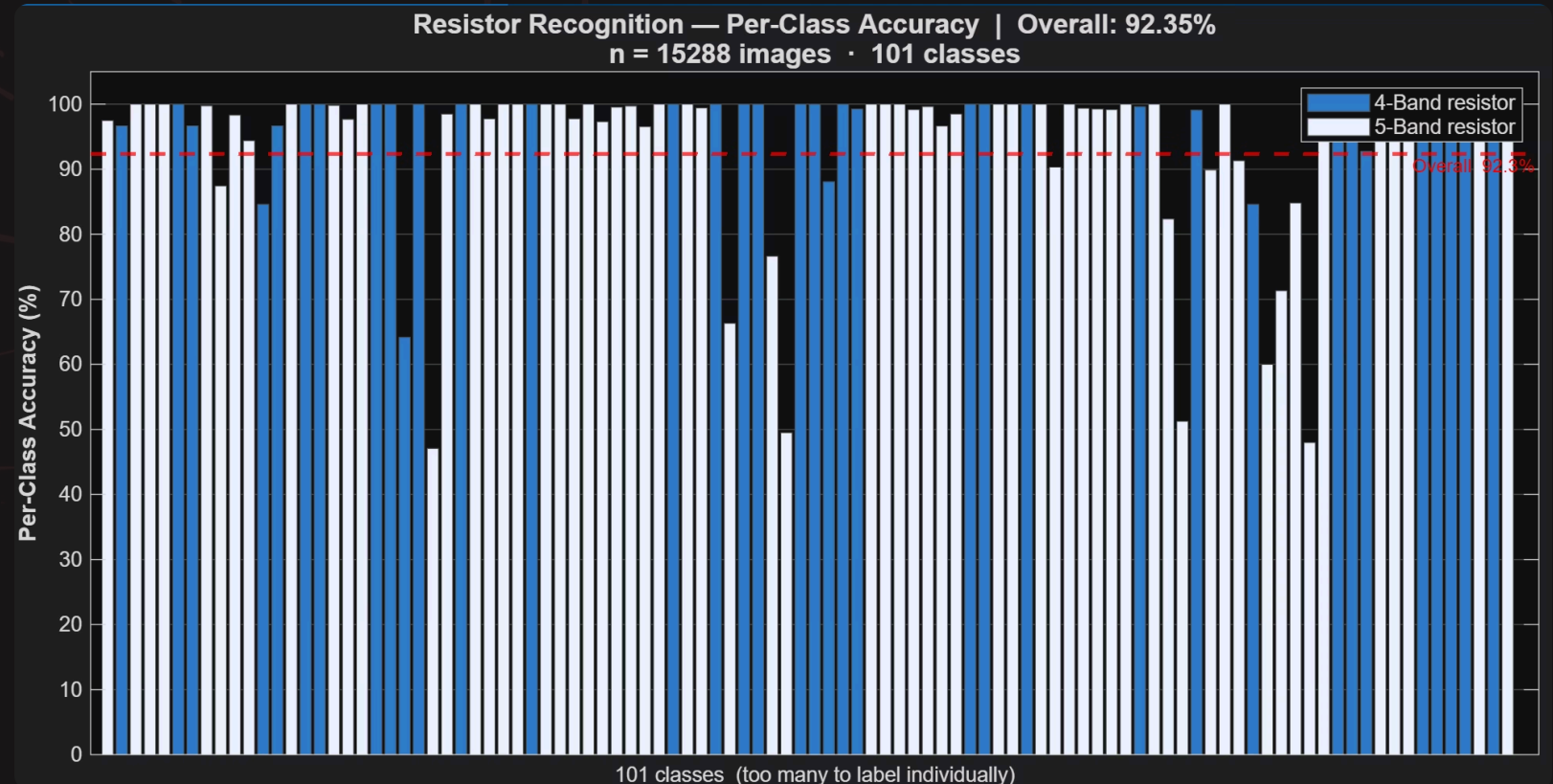
Across 101 resistor classes on 15,288 images

Strong diagonal showing reliable predictions and only minor misclassifications between similar classes.



ResNet 18

Most of ResNet 18's 101 classes hit near-perfect accuracy, but a handful of both 4-band and 5-band resistors dip well below the 92.35% average, revealing specific value combinations the model still struggles to distinguish.



Res Net 18 Obstacles and Solutions

Obstacle:

5-band accuracy was lower (~87%) due to tightly spaced color bands and a small, low-quality dataset.

Solution:

Expanded dataset from 600 → 1,800 images to improve variation and clarity of band patterns.

Impact:

Improved model generalization and classification of complex 5-band resistors.

Obstacle:

(~15,000 images) suffered from severe class imbalance some classes having ~500 images while most had ~50, causing the model to overfit toward the larger classes and degrade overall performance.

Solution:

Introduced a class size limiter so that no class could exceed 50% more samples than the smallest class, effectively rebalancing the dataset.

Impact:

Reduced overfitting and improved generalization across resistor values.

Obstacle:

Image manipulation (hue and contrast adjustments), particularly in the 5-band model, introduced inconsistencies that hurt performance and led to unreliable predictions.

Solution:

Removed these image augmentations from the preprocessing pipeline.

Impact:

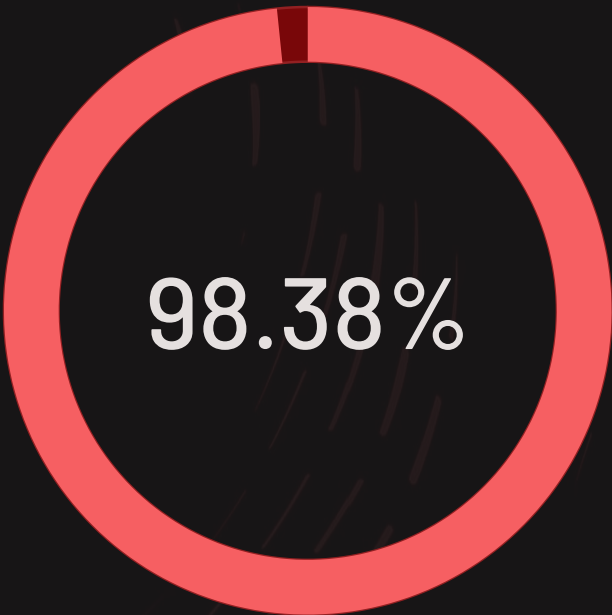
Improved model stability and accuracy by ensuring more consistent input data across all resistor classes.

CNN

CNN (Prototype)



Band Count Classifier
Validation Accuracy



4-Band Value Classifier
Validation Accuracy



5-Band Value Classifier
Validation Accuracy

CNN (Final)



Band Count Classifier
Validation Accuracy



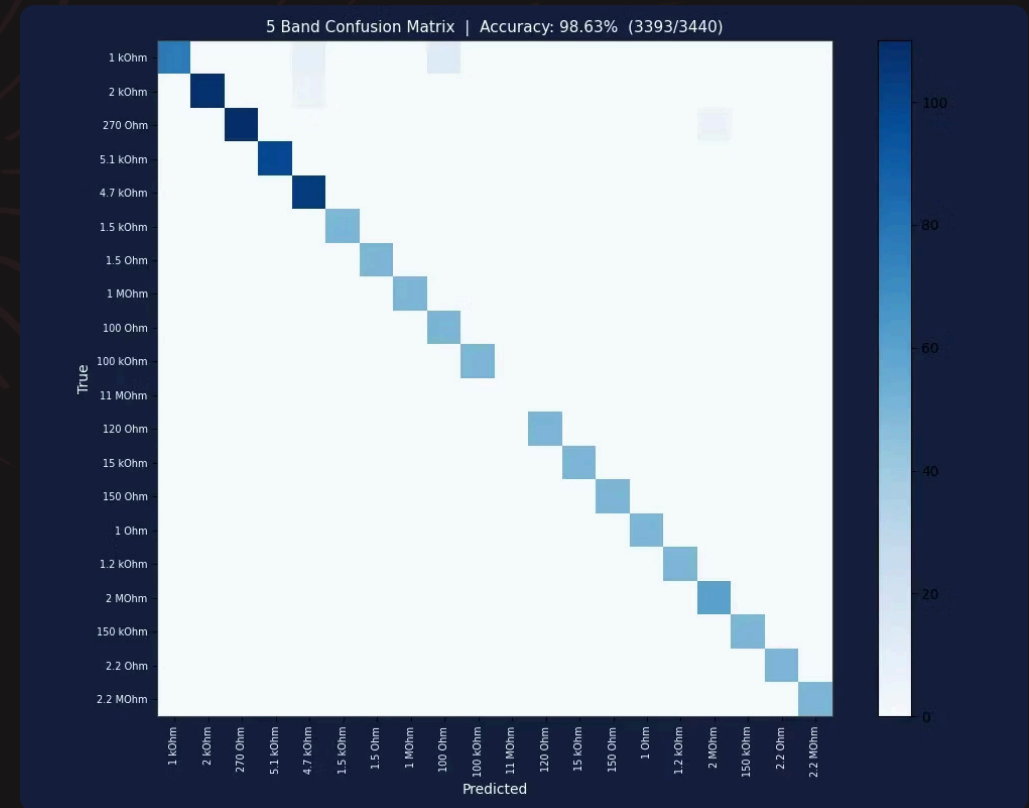
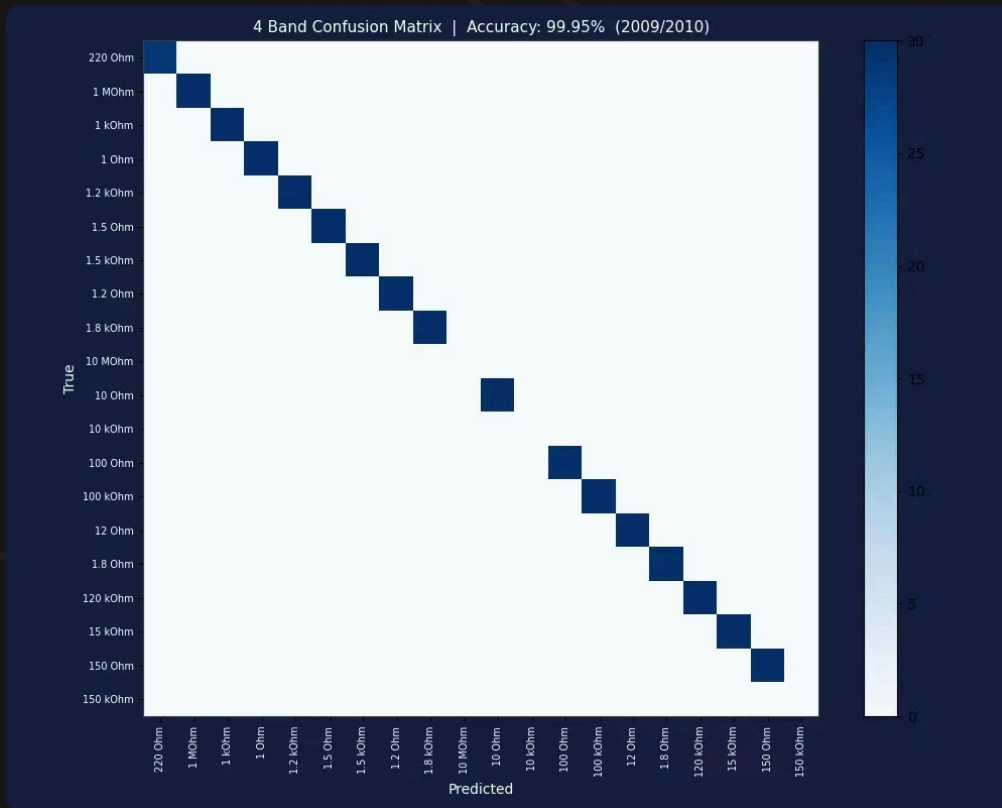
4-Band Value Classifier
Validation Accuracy



5-Band Value Classifier
Validation Accuracy

CNN Performance

When evaluated separately, the model performs near-perfectly, **99.95%** on 4-band and **98.63%** on 5-band resistors with clean diagonals and virtually no off-diagonal errors.



CNN Obstacles & Solutions

1. Data & Input Quality

Camera Resolution

Obstacle: Low resolution (5MP)
limited detail

Solution: Increased to 16MP →
64MP images

Impact: Improved feature clarity
and accuracy

Dataset Parsing

Obstacle: Data spread across
multiple sources

Solution: Combined datasets
into one

Impact: Cleaner pipeline and
consistency

2. Model Design & Training

Model Architecture

Obstacle: Single model
struggled with complexity

Solution: Split into 3 specialized
models

Impact: Improved performance
and efficiency

Training Times

Obstacle: Long training risked
losing progress

Solution: Used checkpoints
during training

Impact: Stabilized training and
preserved progress

Model Architecture

3. Deployment & Optimization

Model Conversion

Obstacle: Model incompatible
with hardware

Solution: Converted .pth → .hef
(Hailo-8L)

Impact: Enabled real-time
inference

Overall Lessons Learned

Resolution Matters

High input resolution caused band colors to bleed (More Usless Data). Downscaling to 128×128 resolved it.

Avoid Color Augmentation

Shifting hue/brightness confused the model, band color is the label. Those transforms were removed.

Diverse Backgrounds Essential

Training on neutral backgrounds then testing on high-contrast ones degraded real-world performance.

Hierarchical Pipeline Wins

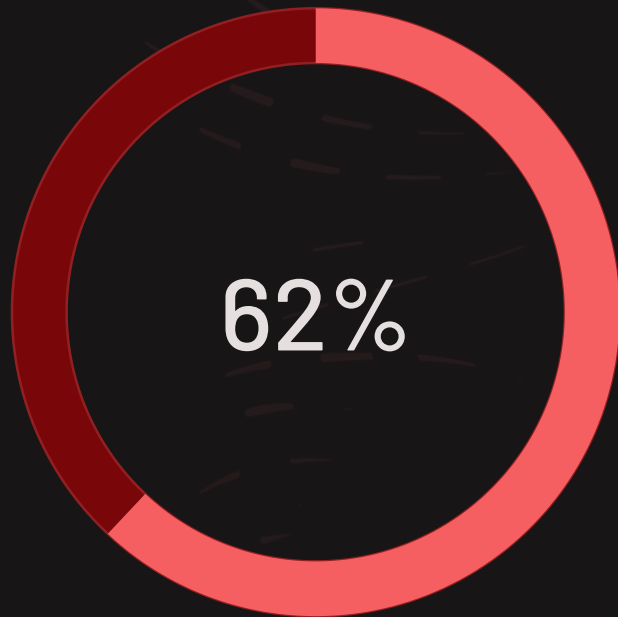
Splitting band count and value into separate classifiers dramatically improved accuracy over a single model.

Real Word comparison

(never before seen image set/ 50 images)

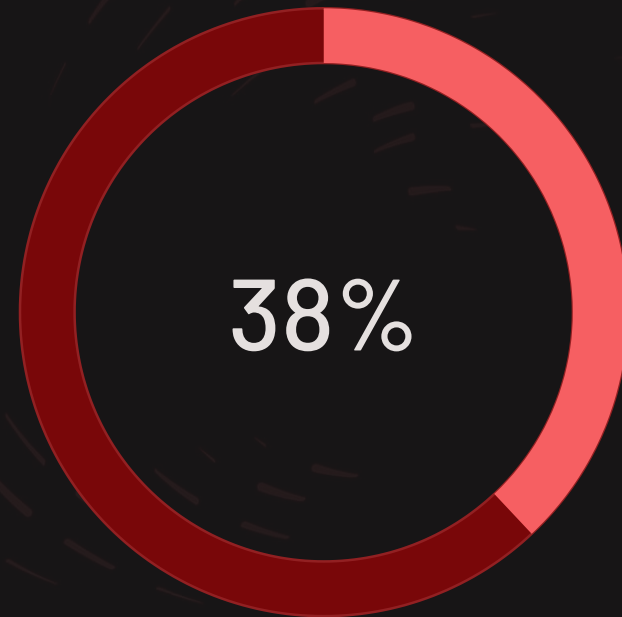
CNN vs ResNet-18 Performance

CNN vs ResNet 18



CNN

Real World Accuracy



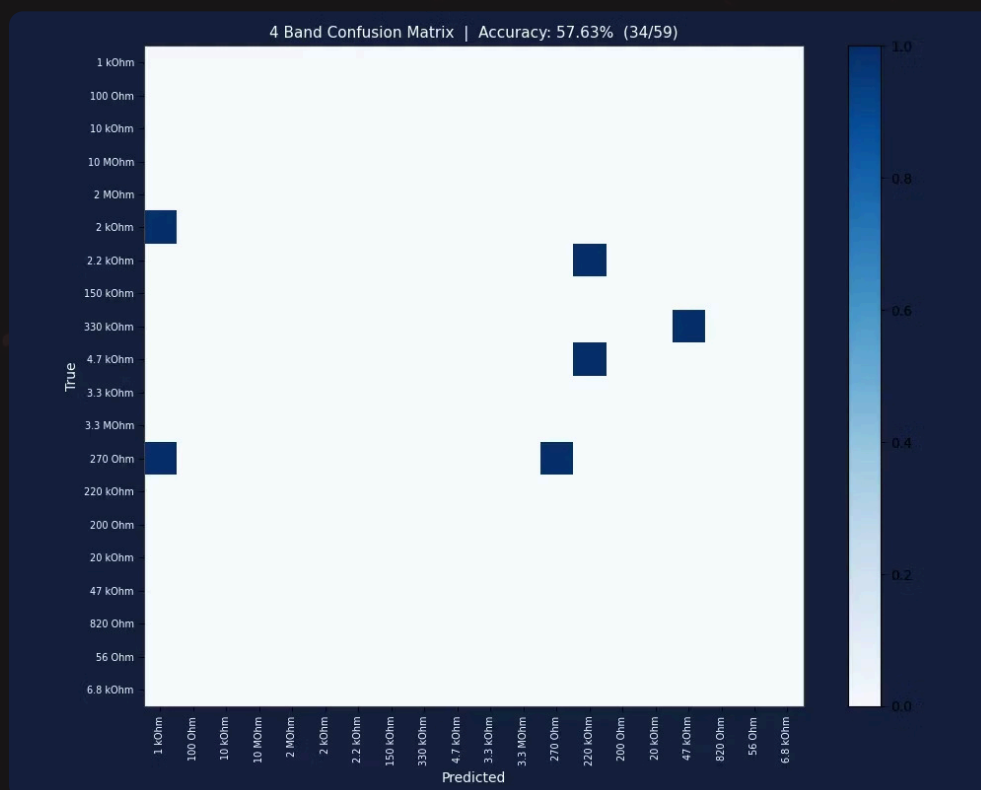
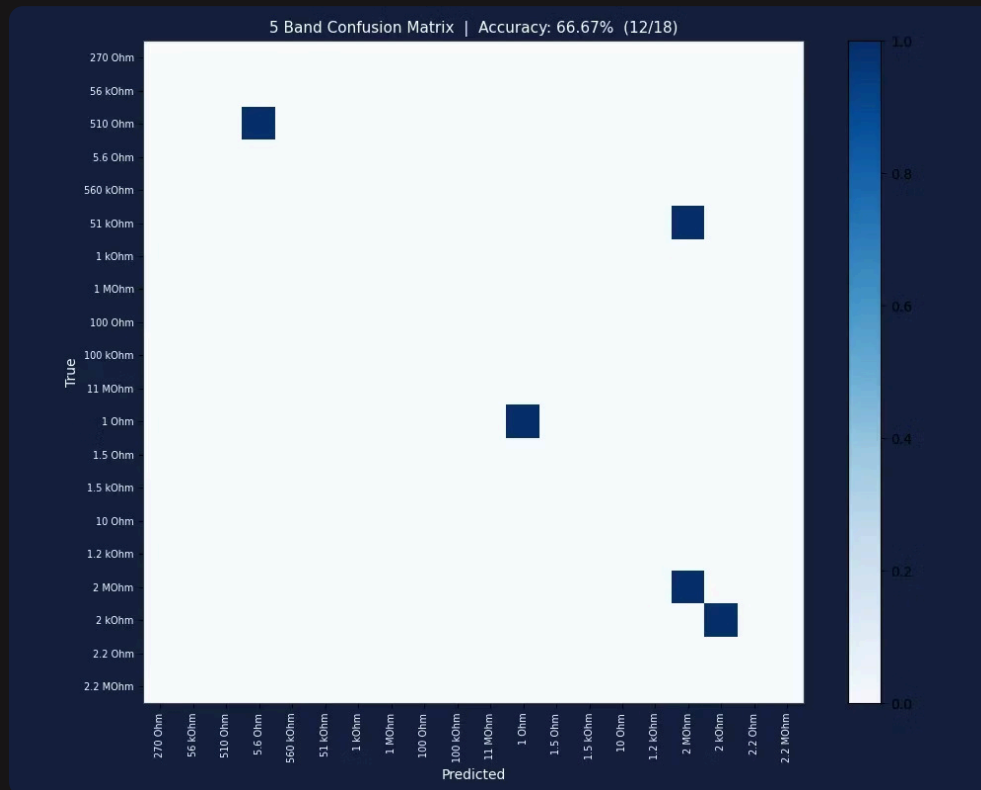
ResNet 18

Real World Accuracy

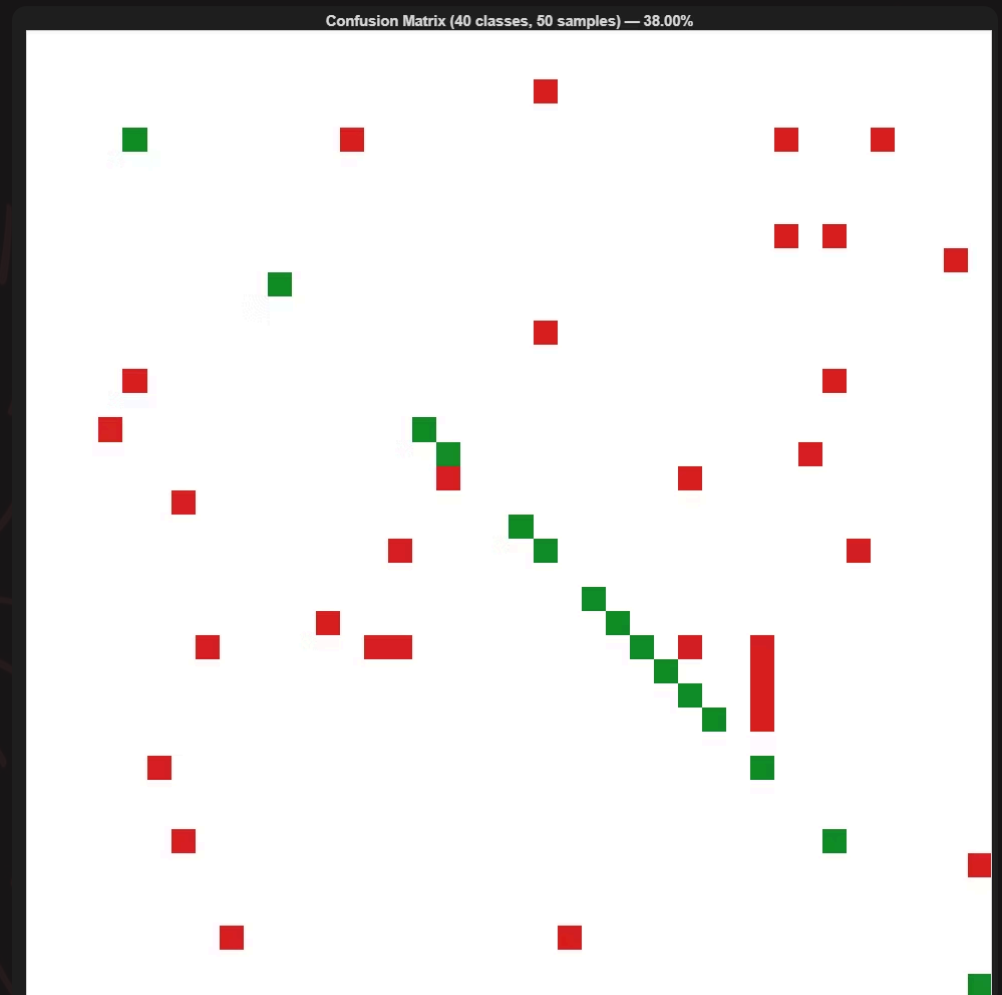
The CNN approach performs substantially better in this real-world resistor classification setting, with a 24-point accuracy advantage over ResNet-18.

Real World Comparison (Raw Data)

CNN Model — 62% Accuracy



ResNet-18 Model — 38% Accuracy



The right plot shows a ResNet-18 model evaluated on a combined 40-class problem (both 4- and 5-band resistors), with an accuracy of 38%.

The left plots show confusion matrices for a CNN-based model evaluated separately on 5-band and 4-band resistors, achieving accuracies of 66.7% and 57.6%, respectively ($\approx 62\%$ average).

Conclusion: CNN vs. ResNet-18

ResNet-18

- Easier to implement and faster to train
- Good baseline performance with limited data
- However, shows **limited scalability** and plateaus due to architectural constraints
- Struggles with complex task

CNN (Custom / Deeper Model)

- Slower training and requires significantly more data
- More complex to tune and optimize
- Achieves **higher overall accuracy and performance**
- Much strong potential for further improvement with additional data and training time

Final Takeaway

- **ResNet-18** is ideal for quick, efficient results
- **CNN** is better suited for maximizing performance when resources allow

Future Work



Larger Dataset

Expand training data with more diverse backgrounds, lighting conditions, and resistor orientations.



Tolerance Classification

Extend the classifier to predict tolerance bands with greater precision across all resistor types.



Mobile Deployment

Optimize the model for real-time inference on mobile devices for field use by engineers and hobbyists.



SMD Resistors

Adapt the pipeline to handle surface-mount (SMD) resistors with numeric codes instead of color bands.

Individual Contributions

Four engineers, two model architectures, one shared goal = accurate resistor recognition.

Ethan Messier

RESNET-18 LEAD

- Led the ResNet-18 training achieving **92.35% overall accuracy (in data set)**
- Built the two-phase fine-tuning strategy with MixUp augmentation
- Made and ran comparative accuracy benchmarks between model versions

Jaleel Khan

RESNET-18 & DATA

- Co-developed and refined the ResNet-18 training pipeline and resistor classifier
- Managed dataset expansion using custom software from **1,270 → 3,800 images** across 89 classes
- Coordinated dataset labeling and class consolidation / fix (90 → 89 classes)

Clay Farley

GUI & HARDWARE LEAD

- Built the GUI using Python
- Designed and built a standalone Raspberry Pi 5 unit for the ECE lab.
- Custom CNN pipeline testing and validation
- Implemented 8L Hailo Accelerator for live image processing

Andrew Rein

CUSTOM CNN IMPLEMENTATION

- Designed the 4-block **VGG-style CNN architecture** from scratch
- Implemented He initialization, Batch Norm, and Global Avg Pooling
- Achieved ~62% **Real world accuracy** on unseen resistors